

Tree of Thought Notebook

Tree of Thoughts (ToT) Notebook — README

Purpose

This notebook implements the **Tree-of-Thoughts (ToT)** reasoning framework to generate reliable and interpretable **sensor recommendations** for underground landmine detection.

For each row in the dataset, the notebook: - Generates **multiple independent reasoning paths** using the Gemma-3-4B-Instruct model

- Extracts **Recommendation, Explanation, and Attention Matrix Analysis (AMA)**
- Scores each explanation on a **1–10 scale**
- Selects the **best-scoring path**
- Saves the final-token AMA as a **JSON file**

The outputs are used later for sparsity analysis, Self-Focus Index (SFI), and cross-method comparisons.

Required Libraries

If running locally, install:

```
pip install transformers accelerate datasets torch pandas graphviz huggingface_hub
```

Hardware requirements: - **GPU strongly recommended** (NVIDIA L4 used in Colab) -

Python 3.10 or later

- Hugging Face account + access token

Model used:

google/gemma-3-4b-it

Required Input Files

Because you will receive only the notebook (not the data), **you must upload the required files to your own Google Drive** before running this notebook.

Required:

File	Purpose	Where YOU must upload it
<code>mines.csv</code>	Dataset of 338 scenarios	Anywhere in your Drive; update path in notebook
AMA output folder	Location where AMA JSONs will be saved	Notebook auto-creates it

Inside the notebook, update the dataset path to whatever location you use, e.g.:

```
df = pd.read_csv("/content/drive/MyDrive/YOUR_FOLDER/mines.csv")
```

How to Run the Notebook (Colab Recommended)

1. Open the notebook in **Google Colab**.
2. Upload `mines.csv` to your own Google Drive.
3. Update the file path in the notebook to match your Drive.
4. Run the cell that installs dependencies.
5. Run the cell that mounts Google Drive.
6. Run the rest of the notebook **top to bottom**.

The notebook will: - Load the Gemma-3-4B-Instruct model
- Process each row
- Save AMA JSONs into a folder on your Drive

- Create the ToT diagram
- Produce a CSV with ToT results

Expected runtime: **1.5–2 hours** on a GPU.

Outputs Generated

1. Final ToT Outputs CSV

A CSV containing:

- `tot_output` — final recommendation + explanation
- `ama_path` — path to its AMA JSON file

You choose where to save it by editing:

```
"/content/drive/MyDrive/YOUR_FOLDER/tot_outputs.csv"
```

2. AMA JSON Files

338 files named like:

```
attn_0000.json  
attn_0001.json  
...  
attn_0337.json
```

Each file stores:

- The final-token attention vector for all transformer layers

The notebook will create the AMA folder automatically, but you can define the location:

```
AMA_DIR = "/content/drive/MyDrive/YOUR_FOLDER/AMA"
```

3. AMA ZIP Files

The notebook splits AMA files into two ZIPs:

```
AMA_part1.zip  
AMA_part2.zip
```

You can customize the save location in the code.

4. ToT Architecture Diagram

Generated as:

```
tot_diagram.png
```

Common Issues & Fixes

Missing or Incorrect Drive Path

Ensure you update paths to match **your** Drive:

```
"/content/drive/MyDrive/..."
```

Hugging Face Login Error

Run:

```
from huggingface_hub import login  
login()
```

Paste your token from:

<https://huggingface.co/settings/tokens>

GPU Memory Issues

Reduce the number of ToT branches:

```
k=3 # default  
k=2 # if needed
```

Summary

This notebook performs: - Multi-branch reasoning (ToT)

- Explanation scoring
 - Sensor selection
 - Attention matrix extraction
 - AMA storage
 - Output CSV creation
 - AMA ZIP archiving
-